

RAK5146 WisLink LPWAN Concentrator

Thank you for choosing **RAK5146 WisLink LPWAN Concentrator** in your awesome IoT project! 🎉 To help you get started, we have provided you with all the necessary documentation for your product.

- [Quick Start Guide](#) 📄
- [Datasheet](#) 📄
- [RAK5146 USB Latest Firmware](#) 📄
- [Quick Start Guide for Raspberry Pi and RAK LPWAN Concentrators](#) 📄

Product Description

The **RAK5146** is an LPWAN Concentrator Module in a mini-PCIe form factor, featuring the Semtech SX1303 and SX126X for the Listen Before Talk (LBT) feature, enabling seamless integration into existing routers or other network equipment with LPWAN Gateway capabilities.

This concentrator can be used with any embedded platform that has an available mini-PCIe slot with SPI/USB connectivity. Additionally, it includes an onboard **ZOE-M8Q GPS** chip.

This module offers an exceptional, comprehensive, and cost-efficient gateway solution. It provides up to 10 programmable parallel demodulation paths, an 8 x 8 channel LoRa packet detector, 8 x SF5-SF12 LoRa demodulators, and 8 x SF5-SF10 LoRa demodulators.

Additionally, RAK5146 can detect an uninterrupted combination of packets at 8 different spreading factors and 10 channels. It can continuously demodulate up to 16 packets, making it suitable for smart metering fixed networks and Internet-of-Things (IoT) applications.

Product Features

- Designed based on **Mini PCI-e form factor**
- **SX1303 baseband processor** emulates 8 x 8 channels LoRa packet detectors, 8 x SF5-SF12 LoRa demodulators, 8 x SF5-SF10 LoRa demodulators, one 125/250/500 kHz high-speed LoRa demodulator, and one (G)FSK demodulator
- 3.3 V **Mini PCI-e**, compatible with **3G/LTE card** of Mini PCI-e type
- Tx power up to 27 dBm, Rx sensitivity down to -139 dBm @ SF12, BW 125 kHz
- Supports **global license-free frequency band**: EU868, EU433, RU864, CN470, US915, AS923, AU915, KR920, and IN865
- Supports optional **SPI/USB** interfaces.
- Listen Before Talk
- Fine Timestamp
- Built-in **ZOE-M8Q** GPS module